Infrastructure as Code

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Outline

Basics

Cloud Provider Tools

Independent Tools

What is Infrastructure?

Virtual Machines

Networks

Databases

Load-balancers

Other cloud resources

Challenge!

How do we manage them?

Managing Infrastructure

Graphical User Interface – click a lot of interesting looking buttons ☺

Command line - write a lot of interesting looking scripts ☺

Potentially many issues

Manual and time-consuming

Complex and error-prone

Scalability

Configuration drift

...

Idea

Express desired infrastructure through code

Infrastructure as Code

Ability to provision and support your computing infrastructure using code instead of manual processes and settings
(`https://aws.amazon.com/what-is/iac/`)

Why code?

Can use best practices from software development

- Version control
- CI/CD pipelines
- Testing
- Documentation

Infrastructure as Code

Key DevOps Practice

Bridges the gap between development and operations

Automate infrastructure setup

Rapid deployment

Repeatable and Consistent deterministic

Easy scalability

...

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IaC in DevOps

Enabling CI/CD by seamlessly integrating into the CI/CD pipelines

Improved testing by creating necessary production-like environments

Support team collaboration as application code and IaC code can be together

Approaches

Imperative

(Task Management)

Developer describes the exact steps to set up the necessary infrastructure

Developer needs to keep track of the system state and manage any necessary changes

Declarative

(State Management)

Developers describes the necessary state of the infrastructure

Automatically handles the necessary updates and changes

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Imperative vs Declarative

Getting Dinner

Imperative	Declarative
What we need to do	What is required
Leave COM2 and go to BIZ2	Go to west coast hawker ctr
Head down Heng Mui Keng Terrace	
Cross the road and take bus 30/51/143	
Get off at west coast hawker ctr	

Imperative	Declarative
Control over every step	Requires less coding skills
Ideal for small tasks	Highly repeatable
	Adapts well to configuration drift
	Starting point does not matter
	Changes to destination handled automatically

Imperative	Declarative
Difficult to maintain	Less control
More prone to human errors	Can over complicate simple tasks
Less idempotent	

Some IaC Tools

Cloud Providers

AWS CloudFormation and CDK

Azure Resource Manager

Google Infrastructure Manager

Independent Tools

Terraform

Pulumi

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AWS CloudFormation and CDK

https://aws.amazon.com/what-is/iac/

imperative

AWS Cloud Development Kit (AWS CDK)

Define cloud application resources with familiar programming languages and interactive configuration tools, all in the IDE

declarative

AWS CloudFormation

Define and manage cloud resources published in the CloudFormation registry, the developer community, and internal libraries

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Azure Resource Manager

https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/overview

Deployment and management service for Azure

Manage infrastructure through declarative templates

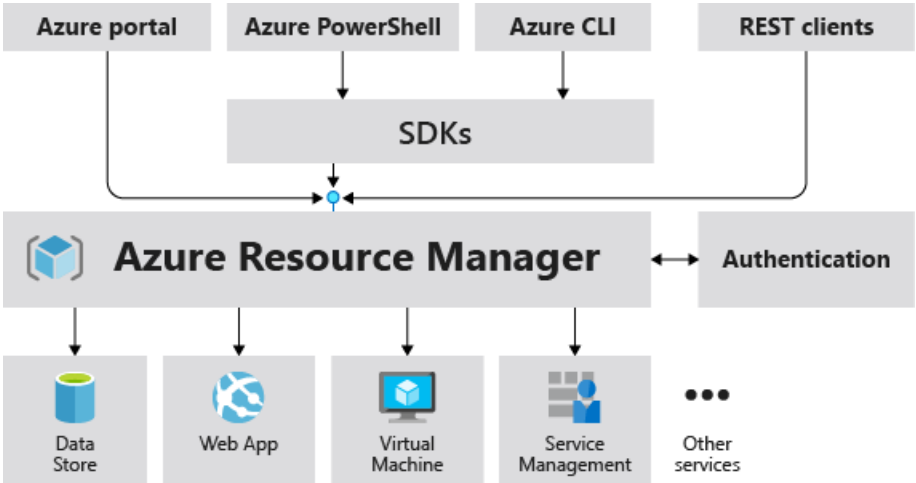
Manage resources as a group

Redeploy resources in a consistent state

Azure RBAC is natively integrated into the management platform

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Azure Resource Manager



Google Infrastructure Manager (Infra Manager)

https://docs.cloud.google.com/infrastructure-manager/docs/overview

New! (Google Cloud Deployment Manager was discontinued on **01/04/26**)

Automates the deployment and management of Google Cloud Infrastructure

Uses . . . Terraform

- Terraform configuration defines the infrastructure
- Configuration is deployed onto Google Cloud by Infra Manager

Application deployment is done via Cloud Build and Cloud Deploy – third party tools can also be used

Pets





Servers are individually managed and cared for

Characteristics

- Named and treated with high importance
- Manual intervention is required for maintenance and recovery

E.g.: Dedicated web-server

Cattle





Servers are interchangeable resources – especially in a cloud environment

Characteristics

- Configured identically
- Easily replaced when they fail, with no need for manual repairs

E.g.: Web-server running (with a load-balancer) that can be quickly changed

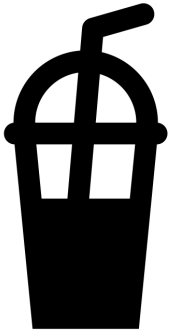
Which one to use?

Recall: IaC is about managing infrastructure

Uses **Cattle** model effectively

Allows for rapid, consistent deployment and management of cloud resources

Take a Break







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Terraform Basics

https://developer.hashicorp.com/terraform

Released in 2014 from HashiCorp

Open-source – though under a Business Source Licence from 2023

Can be used to manage complex, multi-cloud infrastructure

Uses Declarative language

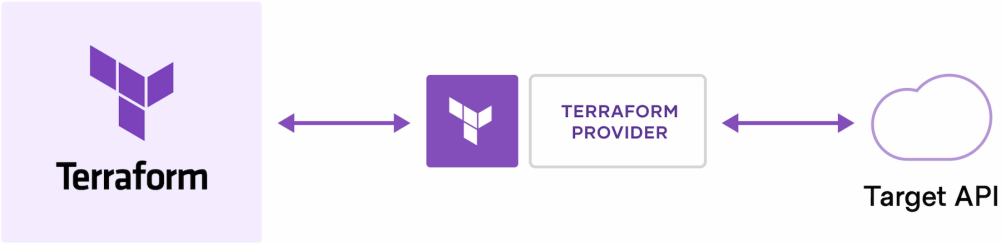
Cloud Agnostic – large list of providers supported

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Terraform

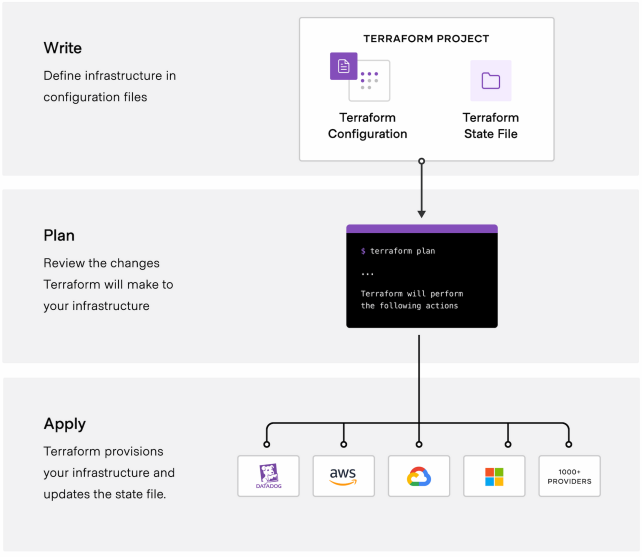
Creates and manages resources on cloud platforms through the use of APIs

Providers enable Terraform to work with virtually any platform or service with an accessible API



Terraform registry contains many thousands of providers – written by HashiCorp and the Terraform community

Core Workflow



Managing Infrastructure

Track Infrastructure

- Keeps track of infrastructure in a state file

Automate changes

- Builds a resource graph to determine resource dependencies and creates or modifies non-dependent resources in parallel

Standardize configurations

- Supports reusable configuration components called modules that define configurable collections of infrastructure
- Encourages best practices
- Can use publicly available modules from the Terraform Registry, or write your own

Collaborate

- Configuration file can be added to a version control system
- Can use HCP Terraform to effectively manage workflow across teams

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Write I

```
# Create repository
$ git init my-infra && cd my-infra

Initialized empty Git repository in /.../my-infra/.git/

# Write initial config
$ vim main.tf

# Initialize Terraform
$ terraform init

Initializing provider plugins...
# ...
Terraform has been successfully initialized!
```

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Write II

```
# Make edits to config
$ vim main.tf

# Review plan
$ terraform plan

# Make additional edits, and repeat
$ vim main.tf
```

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Plan

```
$ git add main.tf
$ git commit -m 'Managing infrastructure as code!'

[main (root-commit) f735520] Managing infrastructure as code!
1 file changed, 1 insertion(+)
```

```
$ terraform apply

An execution plan has been generated and is shown below.
# ...
```

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Apply

```
Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes

# ...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

```
$ git remote add origin https://github.com/*user*/*repo*.git
$ git push origin main
```

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Terraform Language

Main purpose is declaring resources (Infrastructure Objects)

Declarative

Configuration

Document that tells Terraform how to manage a given collection of infrastructure

Can have multiple files and directories

Syntax

Consists of only a few basic elements – Blocks, Arguments, and Expressions

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Example Configuration

Creating an S3 bucket

```
resource "aws_s3_bucket" "example" {
bucket = "my-tf-test-bucket"

tags = {
  Name        = "My bucket"
  Environment = "Dev"
  }
}
```

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Pulumi

https://www.pulumi.com/

Open sourced in 2018

Can write infrastructure code using familiar programming languages – TypeScript, Python, Go, and C#

Flexibility for more complex logic

Available with Agentic AI now ☺

Pulumi Neo - you can now use Natural Language!

Creating an S3 bucket

```
import pulumi
from pulumi_aws import s3

# Create an AWS resource (S3 Bucket)
bucket = s3.Bucket(`my-bucket')

# Export the name of the bucket
pulumi.export(`bucket_name', bucket.id)
```

Learning Curve

Large number of Tools

Code Complexity

Version Control

Configuration Drift configuration drift is reduced, but still not 0, e.g. Git merge conflits

Security and Compliance

Are these enough?

Terraform
Pulumi
OpenTofu
Terragrunt
Spacelift
Puppet
Vagrant
...

`https://spacelift.io/blog/infrastructure-as-code-tools`

Misconfigurations – needs continuous monitoring to ensure no security issues (such as access to sensitive data, incorrect deployment, etc.)

Security – need to protect the code and credentials or access keys, else can lead to data breaches

Compliance – configuration drift may lead to such standards being broken, risking a regulatory violation

Using AI – deploy so and so at so and so with so and so ☺

Policy-as-Code (PaC) – making IaC pipelines policy-aware compliant

Rise of GitOps – automatically deploy changes from Git to system

Self-healing IaC – automatically detect anomalies and update, as necessary

...

Self-healing IaC

Issue

Infrastructure is growing increasingly complex

Three Steps!

Monitor – collect data on workload and current state of servers, etc.

Analyze – use analytics to compare and contrast, use baselines and prediction models as necessary etc.

Remedy – create a remedial plan and action it

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Next Week

Pricing Models & Modeling TCO

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

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